PRODUCT BRIEF

NGen-Realtime

Real-Time Conversational AI System

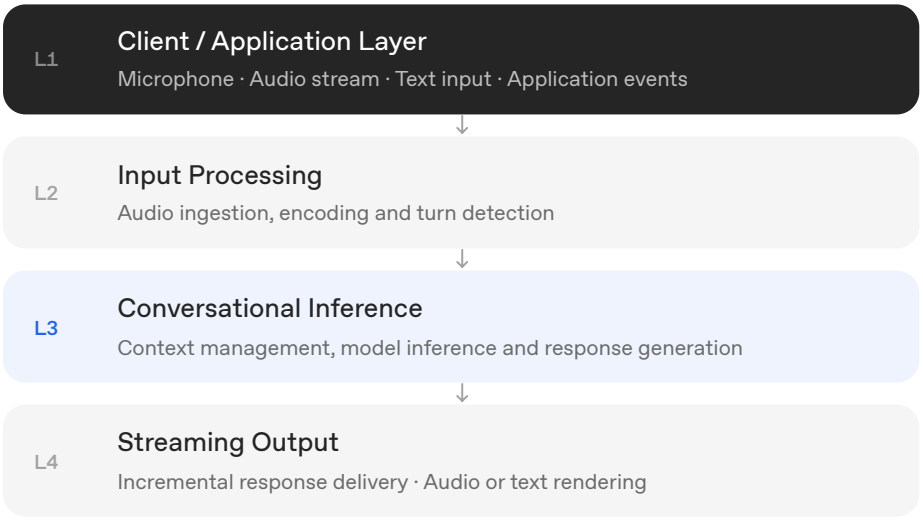
PRODUCT OVERVIEW | SYSTEM ARCHITECTURE | INTERACTION MODEL

01 Overview

NGen-Realtime is TNSA's conversational AI system for building interactive, low-latency AI experiences. It is designed to support communication between users and AI through a real-time interaction pipeline, enabling applications to process incoming conversational input and deliver model-generated responses as the interaction progresses.

Unlike conventional request-response interfaces that treat each exchange as a discrete transaction, a real-time conversational system manages an ongoing interaction. Input processing, model inference, response delivery and conversation state must operate together to maintain continuity.

02 System architecture



NGen-Realtime can be described through four functional layers, with the exact implementation depending on the deployed configuration.

> Conceptual functional architecture. This diagram does not assert that NGen-Realtime uses a native speech-to-speech model or any particular transport protocol.

03 Conversational execution

A real-time interaction follows a continuous execution cycle:

01
Input acquisition
The client captures and transmits the user's supported input.

02
Input interpretation
The system identifies the content and determines when input is ready for processing.

03
Contextual inference
The model processes the available conversational context and generates a response.

04
Incremental delivery
Response content is transmitted to the client as it becomes available, where supported.

05
State continuation
The conversation advances with the relevant interaction history and session state.

Objective Coordinate these stages without introducing unnecessary waiting between the user's input and the system's response.

Runtime capabilities and integration

04 Real-time interaction capabilities

Streaming inference

Incremental output delivery allows client applications to begin rendering a response before the complete response has been generated. This supports more responsive interfaces than workflows that wait for a complete response before displaying or playing it.

Turn management

Conversational turn management determines when user input begins, when it is complete, and when the system should respond. Depending on the implementation, voice activity detection and interruption handling can help coordinate overlapping user and assistant activity.

Conversation state

Session-level state can maintain relevant interaction history, instructions and runtime configuration across successive turns. Context policies determine which prior information is retained and supplied to the model.

Audio processing

Voice-enabled configurations may use either a direct speech-to-speech model or a composed pipeline involving speech recognition, language-model inference and speech synthesis. These approaches differ in latency, component boundaries and control over intermediate representations.

05 Integration model

NGen-Realtime can be integrated into conversational products through the interfaces supported by its deployment.

CLIENT

Client applications

Web, mobile or other supported client interfaces for capturing input, presenting responses and managing user interaction.

TRANSPORT

Streaming transport

A persistent or streaming connection for transferring input, response data and relevant session events, according to the deployed API.

BACKEND

Backend services and tools

Application-side business logic, authorized tool execution and external service integrations where configured.

06 Engineering considerations

Production deployments should be evaluated against measurable system properties:

Latency

Time to first response, time to first audio and end-to-end interaction latency.

Reliability

Connection stability, stream completion, error handling and session recovery.

Security and control

Authentication, access control, protected credentials and appropriate data-handling policies.

Scalability

Concurrent sessions, throughput, resource utilization and cost per interaction.

NGen-Realtime by TNSA

A conversational AI system for developers and enterprises building real-time interactive applications.

TNSA Research
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